

1. "What is the primary goal of data mining?" Choose the Correct Answer

1. What is the primary goal of data mining?
 a) Integrate data b) Clean Data c) Remove redundancy d) Discover Knowledge
2. Which is NOT a step in the Knowledge Discovery Process (KDD)?
 a) Data cleaning b) Data mining c) Data detection d) Data integration
3. Which application benefits LEAST from data mining?
 a) Fraud detection b) Paper filing system c) Web analytics d) DNA sequence analysis
4. When measuring dissimilarity between asymmetric binary attributes, which matches are ignored?
 a) 1-1 matches b) 0-0 matches c) All mismatches d) All matches
5. Why is numeric attribute normalization essential before dissimilarity computation?
 a) Equalize weights b) Make nominal c) remove outliers d) Reduce size
6. Which of these measures data spread?
 a) Mean b) Median c) Mode d) Standard deviation
7. What kind of attribute will you use to define the Rank values?
 a) Interval scaled b) Ratio scaled c) Ordinal d) Nominal
8. Given [4,8,12,15,18,21,24,28,34], what is the smoothed value for 21 using medians (3 equal frequency bins)?
 a) 12 b) 18 c) 21 d) 24
9. What does a covariance of -25 between stock prices indicate?
 a) Prices rise together b) Prices move oppositely c) No relationship d) Perfect correlation
10. Which statement about z-score normalization is false?
 a) Uses mean and SD b) Preserves distribution c) Scales data to [0,1] d) None of them
11. Which test detects redundancy in nominal attributes?
 a) Chi-square b) Correlation coefficient c) Covariance d) Z-score
12. Min-max normalization of \$50,000 (min=\$20,000, max=\$80,000) to [0,1] yields:
 a) 0.25 b) 0.5 c) 0.75 d) 1
13. What is the z-score for value 25 in data with mean=30 and SD=15?
 a) -0.33 b) 0.33 c) 0.5 d) -0.5
14. Which technique would best handle missing "Age" values for Student when "Level" is available?
 a) Delete all records b) Fill with random ages c) Use global mean age d) Use mean age per Level
15. Given the data: {3, 5, 7, 9, 12, 15, 18, 21, 24, 28, 30, 35, 42}, What is the IQR for this data?
 a) 15 b) 18 c) 21 d) 24
16. What are the five numbers represented in a boxplot of this data (in the previous point p15)?
 a) 3, 8, 18, 29, 42 b) 3, 7, 18, 28, 42 c) 3, 9, 18, 30, 42 d) 3, 8, 21, 29, 42
17. In Transactions: {A,B}, {A,C}, {B,D}, {A,B,D}, Given min_support=2, which itemset is frequent?
 a) {A} b) {C} c) {D} d) {A,B}
18. What is the confidence of rule $X \rightarrow Y$ if $\text{support}(X \cup Y)=4$, $\text{support}(X)=8$ and $\text{support}(Y)=4$?
 a) 50% b) 50% c) 75% d) 100%
19. What is the lift of rule $X \rightarrow Y$ if $\text{support}(X \cup Y)=40\%$, $\text{support}(X)=80\%$ and $\text{support}(Y)=40\%$?
 a) 1.0 b) 1.0 c) 1.25 d) 1.5
20. rule $X \rightarrow Y$ has lift=0.8. What does this imply? X and Y are
 a) independent b) negatively correlated c) Positively correlated d) Strongly related
21. Which itemset is pruned by the Apriori property if {A,D} is infrequent?
 a) {D} b) {D} c) {A,B} d) {A,B,D}
22. What is the purpose of reordering items by support in FP-Growth?
 a) minimize prefix sharing b) reduce memory usage c) speed up calculation d) eliminate the pruning
23. If node X's conditional FP-tree is empty, what does this imply?
 a) X is frequent b) min_support is too low c) X is infrequent d) No frequent itemsets contain X
24. If $X \rightarrow Y$ has confidence=45% and min_confidence=60%. What does it mean?
 a) The rule is strong b) The rule is weak c) The items are independent d) The rule should be kept
25. Which metric measures how often itemset X appears in transactions?
 a) Support b) Confidence c) Lift d) χ^2
26. What is the main goal of basket data analysis?
 a) Find size b) Find correlated items c) Predict trends d) Find anomalies

For the given data in table 1. Let Play be the class label attribute. Answer the following points [P27:P34]

27. The entropy of the dataset is:
a) 0.5 b) 0.72 ☒ c) 0.97 d) 1.0
28. The information gain for splitting on "Outlook" is:
a) 0.25 b) 0.32 c) 0.42 d) 0.97
29. If also Info. gain of the other attributes is 0.15, 0.08, 0.22 respectively. The best attribute to split on is:
a) Outlook b) Temperature c) Humidity d) Windy
30. For the rule "IF Windy = No THEN Play = Yes", the coverage is:
a) 30% b) 50% c) 70% d) 90%
31. The accuracy of the rule "IF Outlook = Sunny THEN Play = No" is:
a) 25% b) 50% c) 75% d) 100%
- IF $X = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{Normal}, \text{Windy}=\text{Yes})$: Answer [P32:P35]
32. $P(X|C1)P(C1)$ is:
a) 0.006 b) 0.009 c) 0.015 d) 0.020
33. $P(X|C2)P(C2)$ is:
a) 0.006 b) 0.009 c) 0.015 d) 0.020
34. If you use K-nearest neighbor algorithm with $K = 5$ the prediction of X object will be
a) Play b) Not play c) Can't decide d) Don't know
35. Why is stratified *k*-fold cross-validation preferred over standard *k*-fold?
a) Faster computation b) No test set needed c) Guarantees higher accuracy d) Maintains class ratios in folds
36. Use the given confusion Matrix: The accuracy is
a) 90.5% b) 95.6% c) 96.5% d) 97.8%
37. Which type of outlier deviates based on specific contexts like time or location?
a) Global outlier b) Local outlier c) Collective outlier d) Contextual outlier
38. What defines an outlier?
a) Significant deviation b) Missing data c) Duplicate entry d) Random error
39. Supervised outlier detection suffers from
a) Low computation b) Rare outliers c) High dimensions d) No features
40. If you need to make a prediction you will use
a) K-means b) K-nearest neighbor c) FP-Growth d) DBSCAN
41. Which clustering algorithm initially assumes that all data instance in a single cluster.
a) Agglomerative b) Divisive c) Conceptual d) K-Means
42. Clustering algorithms which can find clusters of arbitrary shape
a) K-means ☒ b) DBSCAN c) Agglomerative d) Divisive
43. Which "V" of Big Data refers to data trustworthiness?
a) Volume b) Velocity c) Veracity d) Value
44. Machine Learning differs from Data Mining by its ability to:
a) Adapt behavior from patterns b) Store large datasets c) Clean noisy data d) Replace human analysts
45. Web Structure Mining analyzes:
a) User activity logs b) Social media content c) Image metadata d) Hyperlink relationships
46. Which chart type displays progress stages?
a) Scatter plot b) Funnel chart c) Box plot d) Pie chart
47. Which characteristic ensures data in a warehouse remains unchanged?
a) Time-variant b) Nonvolatile c) Integrated d) Subject-oriented
48. Rolling up in OLAP means:
a) Aggregating data hierarchically b) Visualizing data c) Drilling into details d) Deleting dimensions
49. In Data warehouse, A star schema consists of:
a) Multiple fact table b) Only metadata c) dimension tables ☒ d) One fact table linked to dimension tables
50. Data Warehousing is characterized by:
a) Real-time processing b) Storing historical data c) Unstructured data only d) Excluding ETL processes

Q2 [10 Marks] Given Points: (2,4), (2,7), (3,5), (4,3), (5,6), (5,2), (6,5), (7,3), (7,7)

1. Assume $k=2$, Initial Centers: $C1=(2,4)$, $C2=(2,7)$ Apply the k-means algorithm using the Manhattan distance and show the new cluster centers after the first-round execution (4 M)

Suppose that your result in (a) is the final cluster, how can you use it to detect the outlier? What is the object which likely be an outlier? (3 M)

If you find the final cluster in (a) what is the silhouette coefficient for object (5,6), what is this value means? (3M)

The end, Best wishes, Amira Rezk

$$Sun = \frac{4}{10} \left(-\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right) +$$

$$Cloudy = \frac{2}{10} \left(0 - \frac{2}{2} \log_2 \left(\frac{2}{2} \right) \right)$$

Table1: for Classification Questions

	222	208	15	32
Play	Windy	Humidity	Temperature	Outlook
No	Yes	Normal	Cool	Rainy
No	No	High	Hot	Sunny
No	No	High	Mild	Sunny
No	Yes	High	Hot	Sunny
Yes	No	High	Hot	Cloudy
Yes	Yes	Normal	Cool	Cloudy
Yes	No	High	Mild	Rainy
Yes	No	Normal	Cool	Rainy
Yes	No	Normal	Mild	Rainy
Yes	No	Normal	Cool	Sunny

confusion matrix

	Predicted: Yes	Predicted: No
Actual: Yes	90 ^{TP}	210 ^{FN}
Actual: No	140 ^{FP}	9560 ^{TN}

$$acc = \frac{TP + TN}{P + N}$$

(3,5), (4,3), (5,3)